

CAPSTONE PROJECT REGISTER

Class: **Duration time: from 11/05/2026 To 06/09/2026**

(*) Profession: <Software Engineer> **Specialty:** <ES> <IS> <JS>

(*) Kinds of person make registers: Lecturer Students

1. Register information for supervisor (if have)

No.	Fullname	Phone	E-Mail	Title
Supervisor	Phạm Minh Trí	0967948487	tripm14@fe.edu.vn	Mr
Supervisor	Đặng Ngọc Minh Đức	0989699299	DucDNM2@fe.edu.vn	Assoc.Pr of

2. Register information for students (if have)

	Full name	Student code	Phone	E-mail	Role in Group

3. Register content of Capstone Project

(*) 3.1. Capstone Project name:

- English: FPT University Research Project Management System
- Vietnamese: Hệ thống quản lý đề tài nghiên cứu khoa học cấp Trường đại học FPT
- Abbreviation: SU26SE053

a. Context:

Managing university-level research projects involves multiple stages, including proposal submission, review, approval, monitoring, and final acceptance. These processes are typically time-consuming, prone to errors, and lack transparency when handled manually. Coordination between faculty members, research staff, administrators, and review committees adds complexity, especially when inviting external experts and organizing online evaluation meetings.

The FPT University Research Project Management System (FURPMS) addresses

these issues by providing a centralized platform to manage the entire research lifecycle, enhanced with AI features powered by LLMs. Projects may last 6 to 12 months, with possible extensions of up to 50% of the original duration (e.g., 9 months for a 6-month project, 18 months for a 12-month project). Key benefits include:

- Generating concise summaries of proposals and reports for quick evaluation.
- Providing AI-driven feedback suggestions based on rubrics and best practices.
- Enabling semantic search across proposals, reports, experts, and reviewer feedback.
- Allowing staff to assign or invite external experts to join review or acceptance committees.
- Supporting online committee meetings via Google Meet or Microsoft Teams integration.
- Offering real-time dashboards, notifications, and analytics to improve transparency and efficiency.

b. Proposed Solutions

Develop an integrated FPT University Research Project Management System (FURPMS) to digitize and automate the management of university research projects.

The system includes:

- Role-based Access Control: Manage roles for Administrator, Staff, Faculty (Principal Investigators), and Review Committees with tailored functionalities.
- Proposal Submission Portal: Enable Principal Investigators (PIs) to submit research proposals with clear guidelines and deadlines.
- Reviewer and Expert Management: Assign internal reviewers or invite external experts to join review or acceptance committees.
- Online Committee Meeting Integration:
 - o Schedule online meetings using Google Meet or Microsoft Teams.
 - o Send automatic invitations with agenda, links, and documents.
 - o Sync with Google Calendar or Microsoft Outlook for seamless scheduling.
- LLM-Powered Features:
 - o Automatic summarization of proposals and reports using an LLM API (e.g., xAI API).
 - o Feedback suggestions for reviewers based on rubrics and best practices.
 - o Semantic search to locate proposals, reports, expert profiles, and feedback by relevant terms.
- Dashboards, Notifications, Analytics, and Reporting Tools: Provide real-time visibility into project statuses, reviewer assignments, and system performance..

● Functional requirement

- User Account Management:
 - o The system shall allow administrators to create, update, delete, and manage user accounts with distinct roles (Administrator, Staff, Faculty, Review Committee).
 - o Role-based access control ensures users access only relevant functionalities.
- Proposal Submission:
 - o The system shall enable Principal Investigators (PIs) to submit research proposals (e.g., PDF, DOCX) with metadata (e.g., title, research area, submission date).
 - o The system shall enforce submission deadlines and provide confirmation of successful uploads.
- Reviewer & Expert Assignment:
 - o The system shall allow staff to manage faculty and expert profiles, assign internal

reviewers, or invite external experts to review or acceptance committees.

- o External experts shall receive automated invitations with secure access to relevant documents.

- **Online Committee Meeting Management:**

- o The system shall integrate with Google Meet or Microsoft Teams APIs to schedule and manage online meetings.

- o Automatic invitations shall include meeting links, agendas, and attached documents.

- o The system shall sync with Google Calendar or Microsoft Outlook for scheduling.

- **LLM Integration:**

- o The system shall integrate an LLM API (e.g., xAI API) to generate concise summaries of proposals and reports, configurable for length and focus.

- o The LLM shall provide feedback suggestions for reviewers based on rubrics and submission content.

- o The system shall support semantic search to locate proposals, reports, expert profiles, and feedback by relevant keywords or phrases.

- **Review & Acceptance Workflow:**

- o The system shall manage the review process, including assigning reviewers, collecting feedback, and finalizing acceptance decisions.

- o Reviewers shall provide feedback and scores based on predefined rubrics.

- **Status Tracking & Notifications:**

- o The system shall provide real-time status updates on proposal submissions, review progress, and acceptance decisions.

- o Automated notifications shall be sent for deadlines, meeting invitations, feedback availability, and acceptance outcomes.

- **Analytics & Reporting:**

- o The system shall generate reports on submission rates, review quality, and project outcomes.

- o Administrators and staff can access analytics on system usage, expert participation, and compliance with university policies.

- **Non-functional requirement:**

N/A

(*) 3.2. Main proposal content (including result and product)

a. **Theory and practice (document):**

The documents include User Requirements, Software Requirement Specification, Architecture Design, Detail Design, System Implementation, Testing Document, Installation Guide, source code, and deployable software packages...

Server-side Technologies:

- o Server: .NET or Java, hosted on cloud platforms like Windows Azure.

- o Database: Relational DBMS (e.g., MS SQL Server, MySQL) for storing assignment, review, and LLM output data, with indexing for search optimization.

- o LLM Processing: Integration with an LLM API (e.g., xAI API) for summarization and feedback suggestions, accessed via Python-based API calls.

Client-side Technologies:

- o Web Client: ReactJS, NextJS, Angular, or any suitable technologies for admin and instructor interfaces.

○ Mobile Client: Flutter, VueJS, or any suitable technologies for student and instructor apps.

b. Products:

1 Web Application for Admin:

1.1 Login

1.2 Logout

1.3 View personal information

1.4 Manage user accounts (Administrator, Staff, Faculty, Review Committee)

1.5 Define research cycles, submission deadlines, and funding allocations

1.6 Oversee proposal submissions and monitor project status

1.7 Generate reports and statistics for university management

2 Web Application for Staff

2.1 Login

2.2 Logout

2.3 View personal information

2.4 Manage faculty and expert profiles, assign reviewers, and invite external experts

2.5 Schedule online meetings with Google Meet/Microsoft Teams integration

2.6 Coordinate proposal submissions and track review progress

2.7 Allocate research resources (e.g., funding, lab access, software licenses)

2.8 Communicate with PIs, reviewers, and committee members via internal messaging

2.9 Review and edit LLM-generated summaries and feedback suggestions

2.10 Perform semantic searches for proposals, reports, and expert profiles

2.11 Manage project extensions and update schedules/deadlines

2.12 Ensure compliance with university policies and generate staff-specific analytics

3 Web/Mobile Application for Faculty (PIs)

3.1 Login

3.2 Logout

3.3 Submit research proposals with metadata

3.4 Update project status and milestones

3.5 Respond to feedback from reviewers and committees

3.6 Submit final reports and research outcomes

3.7 Join online committee meetings via Google Meet/Microsoft Teams

3.8 Search proposals, reports, and feedback using semantic search

3.9 Receive notifications for deadlines, meeting invites, and feedback

4 Web Application for Review Committee

4.1 Accept/decline committee invitations

4.2 Evaluate and review proposals, provide feedback/scores based on guidelines

4.3 Review project completion reports for final acceptance

4.4 View LLM-generated summaries and feedback suggestions

4.5 Join online committee meetings via Google Meet/Microsoft Teams

4.6 Search proposals and feedback using semantic search

4.7 Receive notifications for meeting invites, deadlines, and review assignments

c. Proposed Tasks:

Task package 1: Develop role-based authentication and user management modules, including faculty and expert profile management.

Task package 2: Build proposal submission, deadline tracking, and compliance

monitoring modules.

Task package 3: Implement reviewer assignment, external expert invitation, and compliance workflows.

Task package 4: Integrate Google Meet and Microsoft Teams APIs for online meeting management.

Task package 5: Connect LLM API (e.g., xAI API) for summarization, feedback suggestions, and semantic search.

Task package 6: Create dashboards for Administrator, Staff, Faculty, and Review Committees, including compliance analytics.

Task package 7: Develop mobile/web apps for PIs and committee members, integrating all functionalities.

Task package 8: Build, deploy, and test the system to ensure stability and performance, including extension management.

Task package 9: Prepare all required documents: system analysis and design, test plan, installation manual, user manual, etc. (all students must participate).

Each work group should have many students participating, but there will be one member responsible for the primary responsibility.

Notice: all students are required to understand the reference documents thoroughly and may need to explain them to the viva committee.

4. Other comments (propose all relative things if have).

Supervisor (If have) <i>(Sign and full name)</i> Phạm Minh Trí Đặng Ngọc Minh Đức	HCM, date 09/04/2026 On behalf of Registers <i>(Sign and full name)</i>
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